

TrES-2b

R-1336

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_210605 · created 2026-07-31T21:08:43+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459370.74816 +/- 0.00082 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.092 +/- 0.037
Transit depth (Rp/Rs)^2	0.85 +/- 0.68 [%]
Orbital Inclination (inc)	88.78 +/- 3.06
Airmass coefficient 1 (a1)	1.06 +/- 0.019
Airmass coefficient 2 (a2)	-0.039 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.79 %
Best Comparison Star	#1 - [362, 133]
Optimal Aperture	3.45
Optimal Annulus	6.2
Transit Duration (day)	0.098 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.092 $\pm$ 0.037 vs 0.1254 (deviation 0.75 σ)
Duration fit vs published	0.098 d vs 0.0746 d (deviation 1.02 σ)
Mid-time O-C	-1.8 min
Depth SNR	2.1
Residual scatter	1.79 %

Light curve

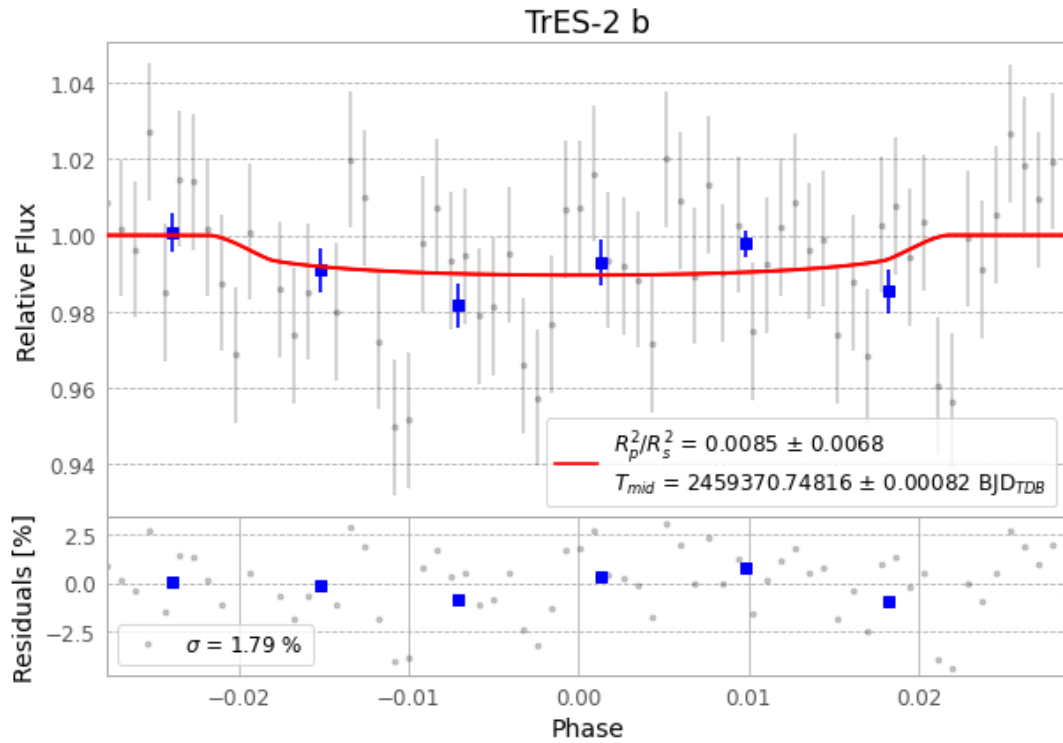


Figure 1 — FinalLightCurve_TrES-2 b_2021-06-04.png (SHA-256 9c4c63b96e31ca4f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [264, 295], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f952d548a0eb3232...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), TRES-2210605041512.FITS → TRES-2210605074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-210605.log (4534c0fec2a7...), FinalLightCurve_TrES-2 b_2021-06-04.png (9c4c63b96e31...), FinalLightCurve_TrES-2 b_2021-06-04.csv (e27800e32a9d...)

Compute resources

Wall time 315.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 21:08Z.